
CRITICAL DATA PROTOCOL: The "Green Light" Rule

TO: All Data Collection Team Members

FROM: Project Lead

SUBJECT: Preventing Data Corruption (Read Before Recording)

1. THE GOLDEN RULE

We are using computer vision (`cv2` + MediaPipe) to train our AI. This system works by detecting your face, hands, and pose and drawing a "Skeleton" over them in real-time.

The Rule is Simple: Look at your screen. If the Green Face Mesh flickers, disappears, or acts glitchy—DO NOT RECORD.

Why? (The "Zero" Problem)

If the camera loses your face for even 1 second, the software fills the data with **Zeros**.

- **Good Data:** `[0.5, 0.4, 0.9...]` (The AI sees eyebrows raised).
- **Bad Data:** `[0.0, 0.0, 0.0...]` (The AI sees nothing).

If we feed "Zeros" to the AI, it will fail to learn the difference between a Question (Eyebrows down) and a Statement (Neutral). **Bad lighting = Broken AI.**

2. THE ENEMY: BACKLIGHTING

90% of detection failures are caused by where you sit in relation to the light.

✗ The Wrong Setup (Backlighting)

- **Do Not:** Sit with a window or bright light **behind** you.
- **Result:** Your camera adjusts to the bright light, making your face a dark silhouette. The AI cannot find your features.
- **Verdict:** unusable Data.

✓ The Correct Setup (Front Lighting)

- **Do:** Sit facing a window or lamp. The light should be hitting your face.
 - **Result:** Your face is evenly lit. The Green Mesh locks on tight.
 - **Verdict:** Perfect Data.
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3. FRAME RATE & SPEED

We are recording frame-by-frame. Some computers are faster than others.

- **Move Normally:** Do not rush the sign. Sign at a natural, conversational pace.
 - **The "Blur" Risk:** If you move your hands too fast in a dark room, the camera creates "motion blur." The AI cannot detect blurry hands.
 - **Solution:** If your room is dim, move slightly slower to help the camera focus.
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4. PRE-FLIGHT CHECKLIST

Before you press "Spacebar" to record, check these 5 things:

1. ☐ **Light Check:** Is the light hitting my face, NOT my back?
2. ☐ **Mesh Check:** Is the Green Face Mesh stable? (Not flickering).
3. ☐ **Hand Check:** Put your hands up. Does the AI draw the pink/red skeleton on them instantly?
4. ☐ **Distance:** Am I close enough that my face and hands are clearly visible, but far enough that my hands don't go off-screen when I stretch my arms?

5. ☐ **Background:** Is the room behind me relatively tidy? (A messy background can sometimes confuse the AI).
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REMINDER: It is better to delay recording by 1 hour to fix a lamp than to record 100 videos that act as "poison" to our model.

If in doubt, send a screenshot of your setup to the Project Lead.